WHITE PAPER

AI-Augmented Instructional Design

A practical framework for using AI across the ADDIE process — to move faster without lowering the bar on learning quality.

ANALYSIS → DESIGN → DEVELOP → IMPLEMENT → EVALUATE

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What's inside

A stage-by-stage guide to applying AI across ADDIE — with real tools, real prompts, and the guardrails that keep quality intact.

SECTION 01

Why AI, why now

Instructional design has always been a quality discipline wrapped around a slow process. AI doesn't change the discipline — it changes the speed of everything around it.

For decades, the bottleneck in instructional design wasn't ideas — it was hours. Transcribing a 90-minute SME interview took 90 minutes. Storyboarding a 20-screen module took a day. Recording, re-recording, and editing narration for a single course could eat an entire week. None of that time made the learning experience better. It was just the cost of getting from concept to course.

Generative AI collapses that cost. The same SME interview that took 90 minutes to transcribe and summarize can be processed in under five. A first-draft storyboard that used to take a day can be outlined in an hour, leaving the rest of the day for the part that actually matters: deciding whether the sequencing, pacing, and practice opportunities are right for the learner.

> **THE CORE IDEA**
>
> AI should absorb the mechanical work of instructional design — transcription, first drafts, formatting, rough narration — so your team's time goes toward judgment: is this the right objective, the right scenario, the right level of challenge for this learner?

What this paper covers

This is not a tool review. It's a process guide — organized around the ADDIE model your team already uses — that shows specifically where AI tools fit, what to ask them for, and where a human still has to make the call. Every section includes real tools, example prompts, and a quality checkpoint.

60-70%	5 stages	0
Typical time reduction reported on first-draft content tasks (scripts, storyboards, summaries)	Of ADDIE where AI has a distinct, well-defined role	Stages where AI should make the final quality call alone

Time-savings figures are directional estimates based on common AI-assisted ID workflows reported by practitioners; actual results vary by project complexity and tool fluency.

SECTION 02

The AI-augmented ADDIE framework

Same five phases. Different center of gravity — AI handles the first draft, you handle the judgment.

PHASE	WHERE AI HELPS	HUMAN STAYS IN CONTROL OF
Analysis	Transcribing & summarizing SME interviews, drafting needs analysis from notes, surfacing themes across stakeholder input	Which performance gap is the real problem; what's worth solving
Design	First-draft design documents, learning objectives, scenario scripts, storyboard shells, evaluation plan outlines	Sequencing, objective alignment, scenario realism, assessment validity
Development	Course imagery, rough-cut video, narration audio, copy editing & tone passes	Visual accuracy, brand fit, narration pacing, final copy approval
Implementation	FAQ drafts, facilitator guide first passes, support copy	Rollout plan, stakeholder communication, LMS configuration
Evaluation	Survey drafting, open-text response summarization, draft evaluation reports	Interpreting results, deciding what changes next

A simple rule of thumb

> *If a task is mechanical and reversible — a first draft, a transcript, a rough cut — let AI do it. If a task requires judgment about the learner — is this the right scenario, the right tone, the right level of challenge — that decision stays with you.*

The rest of this paper walks through each phase in order, with the specific tools, prompts, and quality checks that make this rule of thumb practical instead of aspirational.

SECTION 03 — ANALYSIS

SME interviews & needs analysis

The analysis phase generates the most raw, unstructured input of any ADDIE stage — and it's where AI delivers the fastest, lowest-risk win.

Capturing & summarizing SME interviews

Every SME interview produces far more material than makes it into your needs analysis. The old workflow — listen back, pause, type, repeat — is where hours disappear. AI transcription and summarization tools compress that into minutes, and let you stay present in the conversation instead of scribbling notes.

Otter.ai

Real-time transcription with speaker labels — ideal for live or virtual SME interviews. Auto-generates a summary and action items after the call.

Fireflies.ai

Joins your Zoom/Teams call automatically, transcribes, and lets you ask questions of the transcript afterward — useful for pulling specific quotes for a design document.

Notion AI

Paste a raw transcript in and prompt it to extract themes, pain points, and candidate learning objectives directly into your project workspace.

> EXAMPLE PROMPT
>
> "Here is a transcript of an interview with a subject matter expert on [topic]. Summarize the three to five recurring performance problems they describe, and draft a list of candidate learning objectives that would address each one."

Drafting the needs analysis

Once you have clean, summarized interview data, AI can produce a strong first-draft needs analysis — structured, organized, and ready for your edit pass, rather than a blank page.

- **Feed it structure, not just notes.** Give the model your standard needs analysis template (the gap, the audience, the constraints, the recommended solution) and ask it to populate each section from the interview summary.

- **Ask for gaps, not just answers.** A strong prompt asks the model to flag what's still unclear — open questions you should follow up on with the SME — not just to fill in everything confidently.
- **Treat the output as a draft, always.** AI is good at organizing what it's given. It cannot tell you whether the SME described the real problem or a symptom of it — that's still a human read.

SECTION 04 — DESIGN

Artifacts, storyboards & scenarios

The design phase produces the documents that align your whole project — and AI is exceptionally good at producing strong first drafts of every one of them.

Key artifacts AI can first-draft

ARTIFACT	WHAT TO FEED THE MODEL
Design document	Needs analysis summary + your standard design doc template structure
Learning objectives	The performance gap + audience + desired behavior change (ask for Bloom's-aligned action verbs)
Evaluation plan	Learning objectives + your Kirkpatrick or other evaluation framework
Storyboard shell	Learning objectives + module length + your storyboard template columns
Scenario scripts	A real workplace situation from the SME interview + desired decision points

EXAMPLE PROMPT — SCENARIO SCRIPT

"Write a branching scenario script for a new manager learning to handle a performance conversation. The learner plays the manager. Include three decision points, each with one correct, one partially correct, and one poor choice, with brief feedback for each. Base the situation on this SME description: [paste]."

Why scenario work is a strong AI use case

Realistic workplace scenarios require lots of plausible dialogue and branching paths — exactly the kind of high-volume, structured generation AI is suited for. You provide the real situation and the decision points that matter; AI drafts the dialogue and branch text for you to tighten.

Where design judgment still has to lead

- ✓ Does this scenario reflect a real, common situation — or a contrived one that doesn't ring true to your learners?
- ✓ Are the objectives actually measurable, or just AI-generated language that sounds like an objective?
- ✓ Does the evaluation plan match how your organization actually measures performance change?

ChatGPT (GPT-4 class models)

Strong general-purpose drafting for design docs, objectives, and scenario scripts. Works well with a reusable custom prompt or saved template.

ChatID — Instructional Design GPT

IDC's own GPT, purpose-built for ID artifacts: needs analyses, learning objectives, design documents, and project plans, with ID-specific structure built in.

Claude

Particularly strong for longer design documents and evaluation plans where structure and internal consistency across a long document matter.

SECTION 04 — DESIGN, CONTINUED

A worked example: storyboard shell

Below is a simplified example of what an AI-drafted storyboard shell looks like before a designer's edit pass — useful as a reference for what to expect from a first draft.

SCREEN	ON-SCREEN TEXT	NARRATION (DRAFT)	NOTES / MEDIA
1.1	Welcome & objective	"By the end of this module, you'll be able to identify the three warning signs of..."	[Draft — confirm objective wording matches design doc]
1.2	Scenario intro	"Meet Jordan. Jordan just received a complaint from a client about..."	[Draft — needs SME-reviewed scenario detail]
1.3	Decision point 1	"What should Jordan do first?"	[AI-suggested 3 options — needs realism check]

This is a draft shell — every cell is a starting point for the designer's review, not a final asset.

How to prompt for this output

EXAMPLE PROMPT

"Using this storyboard template [paste columns: screen #, on-screen text, narration, media notes], draft screens 1 through 5 for a module on [topic]. Base the scenario on [SME summary]. Keep narration to 2-3 sentences per screen, conversational tone, second person."

SECTION 05 — DEVELOPMENT

Images, video & narration

Media production has historically been the most expensive and time-consuming part of course development. AI tools now generate strong first-pass media directly from a storyboard.

Course imagery

Custom illustrations and photos used to mean a stock photo budget or a designer with hours to spare. AI image generation produces on-brand visuals directly from a text description — useful for custom diagrams, scenario illustrations, and concept art that stock libraries don't have.

Midjourney
Best-in-class for stylized illustration and concept art. Strong for scenario characters and custom visual metaphors.

Adobe Firefly
Commercially safe (trained on licensed content) and integrates directly into Photoshop/Express — a strong choice when IP clearance matters for client work.

DALL·E 3 (via ChatGPT)
Good for quick diagram-style and conceptual imagery generated in the same conversation as your script or storyboard.

Course video

AI video tools now generate full scenario videos — including a presenter — directly from a script, without a camera, studio, or actor.

Synthesia
AI avatars deliver your script on camera in 140+ languages. Strong fit for compliance training, software walkthroughs, and scenario role-plays where a real shoot isn't practical.

HeyGen
Similar avatar-based video generation with strong lip-sync quality — good for shorter microlearning videos and multilingual rollouts.

Descript

Edits video by editing the transcript — remove filler words, restructure a take, or generate an "overdub" voice correction without re-recording.

Narration audio

AI voice generation has reached a quality level where it's appropriate for a meaningful share of corporate training narration — particularly for draft review cycles and lower-stakes content.

ElevenLabs

The current standard for natural-sounding AI narration. Supports voice cloning (with consent) so your course can keep a consistent "house voice" across modules.

Murf AI

Built specifically for e-learning and corporate narration, with pacing and emphasis controls tuned for instructional content.

PlayHT

Strong multilingual narration — useful when the same course needs to roll out in several languages without re-recording with human voice talent each time.

PRACTICAL WORKFLOW

Use AI narration for your review-cycle drafts even if you plan to record human narration for the final release. Stakeholders can react to pacing, tone, and timing days earlier — before a studio session is ever booked.

SECTION 05 — DEVELOPMENT, CONTINUED

The AI-assisted media pipeline

A practical sequence for taking an approved storyboard to a finished media-rich module — with the human checkpoint marked at each handoff.

STEP	AI DOES	TOOL EXAMPLE	HUMAN CHECKPOINT
1	Generate narration script audio (draft)	ElevenLabs / Murf	Pacing & tone match brand voice?
2	Generate on-screen visuals from storyboard notes	Firefly / Midjourney	Accurate, on-brand, no IP issues?
3	Generate presenter / scenario video	Synthesia / HeyGen	Lip-sync, framing, realism acceptable?
4	Edit & assemble rough cut	Descript	Final pacing, transitions, accessibility (captions)
5	—	Designer/SME review	**Final sign-off before publish**

A NOTE ON ACCESSIBILITY

AI-generated video and narration still need human-verified closed captions, alt text for AI-generated images, and a check that any avatar-based video meets your organization's accessibility standards. AI tools are getting better at this natively — but treat it as a checklist item, not an assumption.

SECTION 06 — DEVELOPMENT

Copy editing & review

AI is an exceptionally good line editor — and a fast way to get a second set of eyes on course copy before it goes to a SME or stakeholder review.

What AI catches well

- ✓ Reading-level mismatches between your copy and your stated audience
- ✓ Inconsistent terminology across screens (e.g. "manager" vs. "supervisor" used interchangeably)
- ✓ Passive voice and wordiness that slows down on-screen reading
- ✓ Tone drift — copy that starts conversational and turns formal mid-module
- ✓ Accessibility issues in phrasing (e.g. directional language like "click the button on the right" that assumes sighted navigation)

What still needs a human read

- ✓ Whether the content is factually correct for your specific organizational context
- ✓ Whether the tone matches your brand voice in ways a style guide can't fully capture
- ✓ Whether a scenario or example will land the right way with your specific audience

Grammarly Business

Reading-level, tone, and clarity scoring built into your authoring workflow — useful as a first-pass filter before a human edit.

Hemingway Editor

Flags overly complex sentences and passive voice — a quick way to tighten course copy for readability.

Claude / ChatGPT

Paste in a screen of course copy and a style guide; ask for a tracked-changes-style edit pass with reasoning for each change, not just a rewrite.

EXAMPLE PROMPT

> "Review this course copy against our style guide [paste]. Flag any terminology inconsistencies, passive voice, and reading level above a 9th-grade level. For each issue, suggest a specific edit and explain why."

SECTION 07

Modern authoring tools with built-in AI

Beyond standalone AI tools, most major e-learning authoring platforms now build generative AI directly into the course-building workflow.

Articulate Rise AI

Generates a complete first-draft course — lessons, content blocks, and knowledge checks — from a topic prompt or outline, directly inside Rise.

Articulate Storyline AI Assistant

AI-assisted slide text generation, translation, and image generation built into the Storyline 360 authoring environment.

Articulate AI features (360 suite)

Across the 360 suite — auto-generated quiz questions, alt text suggestions, and translation tied directly to your source content.

Adobe Captivate

AI-powered "Text-to-Speech" and generative slide creation, plus Firefly integration for course imagery directly inside the tool.

Synthesia Course Creator

Goes beyond avatar video — generates a structured course outline and matching video content from a single prompt or source document.

Gomo Learning

AI-assisted content generation and translation built into a cloud-based, responsive authoring environment.

HOW TO EVALUATE A TOOL'S "BUILT-IN AI"

Ask three questions: (1) Can I see and edit the AI's reasoning or just the output? (2) Does it use my organization's content as a source, or only its general training data? (3) What happens to the data I upload — is it used to train the vendor's models? Get a clear answer on all three before rolling a tool out broadly.

SECTION 08

Quality guardrails — keeping the bar high

Speed without a quality framework just produces mediocre courses faster. These five guardrails keep AI-assisted work at the same bar as your best human-only work.

1. Every AI output gets a named human reviewer

Not "the team reviews it" — a specific person, accountable for sign-off, on every AI-generated artifact before it moves to the next ADDIE phase.

2. AI drafts the first 80%, never the last 20%

Final tone calibration, factual verification with the SME, and the decision that a scenario is realistic enough — these stay human, every time, regardless of how strong the draft looks.

3. Fact-check anything AI generates about your specific content

AI is fluent, not necessarily accurate — particularly about your organization's specific policies, products, or procedures. Treat any specific claim in AI-generated content as unverified until a SME confirms it.

4. Keep a consistent prompt library, not ad hoc prompting

Save your best-performing prompts for each artifact type (needs analysis, storyboard, scenario script) as a shared team resource. This keeps quality consistent across designers and projects, the same way a template does.

5. Disclose AI use where it matters

If a course uses an AI avatar presenter or AI-generated voice, many organizations choose to disclose this to learners — both as a trust practice and, increasingly, as a compliance consideration in certain industries and regions.

> **THE LITMUS TEST**
>
> Before publishing anything AI-assisted, ask: "If a learner found out this was AI-generated, would they trust it less?" If the honest answer is yes, that's a signal the human review pass needs another round.

SECTION 09

Getting started — a 30-day rollout plan

You don't need to overhaul your whole process at once. Here's a low-risk sequence for introducing AI across one real project.

WEEK	FOCUS	ACTION
Week 1	Analysis	Pilot AI transcription/summarization on your next SME interview. Compare the AI summary against your own notes.
Week 2	Design	Draft one design document and one storyboard shell with AI assistance. Time how long the edit pass takes vs. writing from scratch.
Week 3	Development	Generate draft narration audio and one piece of course imagery with AI. Run it past your usual review process.
Week 4	Review & decide	Meet as a team. Decide which AI steps earned a permanent place in your workflow — and where the draft quality wasn't worth the prompt time.

Signs it's working

- ✓ Your team is spending more time on sequencing, scenario quality, and feedback design — not less time on quality overall
- ✓ Edit cycles are getting shorter because the first draft is stronger, not because review is getting skipped
- ✓ SMEs report the design documents reflect their input more accurately, not less

Signs to slow down

- ✓ Reviewers are rubber-stamping AI drafts instead of genuinely reviewing them
- ✓ Scenario content is starting to feel generic or interchangeable across projects
- ✓ Nobody can explain why a specific AI-generated claim made it into the final course

SECTION 10

Resources & further reading

Tools and templates referenced in this paper, in one place.

Analysis & transcription

- Otter.ai — real-time transcription & summarization
- Fireflies.ai — meeting transcription with searchable Q&A
- Notion AI — theme extraction & drafting workspace

Design & artifacts

- ChatID — Instructional Design GPT (IDC)
- ChatGPT · Claude — general-purpose drafting
- IDC Instructional Design Templates — needs analysis, design doc, storyboard templates

Media production

- Midjourney · Adobe Firefly — imagery
- Synthesia · HeyGen — AI presenter video
- Descript — video & audio editing
- ElevenLabs · Murf AI · PlayHT — narration

Copy editing

- Grammarly Business · Hemingway Editor

Authoring platforms with built-in AI

- Articulate 360 (Rise & Storyline)
- Adobe Captivate
- Gomo Learning

Keep learning with IDC

- IDC Blog — practical articles on ID, AI, and career growth
- IDC Needs Analysis Tool
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